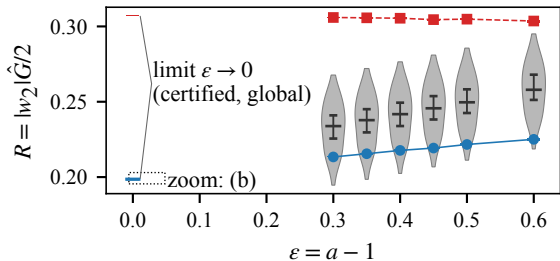
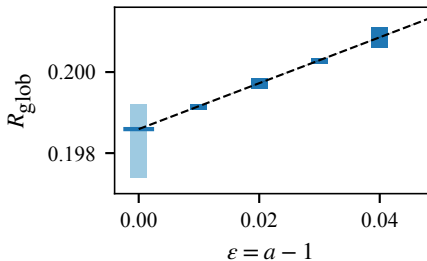


(a) certified thresholds and free-training crossings



(b) small  $\varepsilon$  and the registered  $c_1$  test



(a) free-training crossings,  $n = 38$  runs per  $a$  (violins; medians with bootstrap 95%); boxes: certified intervals to scale; limit  $R_{\text{solve}}^{\infty}$  global.

(b)  $n = 4$  certified brackets ( $a = 1.01\text{--}1.04$ ) and the limit (no runs); unconditional B&B bracket of  $R_{\text{glob}}^{\infty}$  and sharp value  $[0.1985926, 0.1985927]$ ; first order  $R^{\infty}(1 + c_1\varepsilon)$ ,  $c_1$  certified; first-order range certified for  $|\varepsilon| \leq 0.05$ ; registered  $c_1$  test: INCONCLUSIVE: feasible  $c_1 \in [0.225, 0.330]$ , width  $0.105 > 0.1$ .